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Enhanced mechanical properties of a Fe-Mn-Al-C austenitic low-density steel by increasing hot-rolling reduction

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ABSTRACT

The microstructure evolution and mechanical properties of a Fe-Mn-Al-C austenitic low-density steel at different hot-rolling reductions have been investigated. It was found that with the increase in hot-rolling reduction from 45% to 82%, the grain size was uniformly refined and the dislocation density was continuously increased. As a result, both yield strength and ultimate tensile strength were effectively enhanced while the ductility and toughness only slightly decreased. An excellent combination of yield strength, ultimate tensile strength, total elongation, and toughness of 648 MPa, 976 MPa, 50%, and 105 J/m^2, respectively, was obtained. The yield strength increment was interpreted by the stronger grain boundary strengthening and dislocation strengthening, and the minor loss of ductility and toughness was arising from the faster saturation of the slip band refinement (slip band spacing reached ~50 nm) and an earlier occurrence of microbands due to the finer grain size. Based on this research, it is proposed that increasing the hot-rolling reduction was an effective approach to enhance the overall mechanical properties of the low-density steel with a fully austenitic microstructure.

1. Introduction

In response to the requirement for green manufacturing concepts such as lightweight design, energy-saving, and carbon emission reduction, the development of high-performance steels with low density was in urgent demand. In this regard, the Fe-Mn-Al-C low-density steels with excellent weight-reduction potential and extraordinary mechanical properties have become an emerging class of structural material in the automotive, aircraft, and ship industries [1,2]. The effective density reduction up to ~18% of Fe-Mn-Al-C low-density steels compared to the traditional advanced high-strength steels, e.g., dual-phase (DP) steels, transformation-induced plasticity (TRIP) and twinning-induced plasticity (TWIP) steels, etc., was realized by alloying with a high amount of light elements such as Al and C [3]. Depending on the relative alloying element content and microstructure constitutes, the Fe-Mn-Al-C low-density steels can be classified into three categories: ferritic steel, austenitic steel, and duplex steel [4–7], among which the austenitic low-density steel covering the chemical composition range of (15–30%)Mn-(6–12%)Al-(0.5–1.2%)C (wt%) exhibits the most attractive mechanical properties with the yield strength (YS) over 500 MPa, ultimate tensile strength (UTS) over 800 MPa, and total elongation (TE) over 40% [7–9].

The outstanding strengthening effect of the austenitic low-density steel is arising from the homogeneously distributed nano-sized (Fe, Mn)$_3$AlC$_x$ carbides (κ-carbides) within the austenite matrix. The intragranular κ-carbides are generally precipitated via aging treatment in the temperature range of 450–700 °C [10,11]. Nevertheless, with increasing aging time, the strengthening effect of κ-carbides will approach the limit due to the coarsening of κ-carbides, while the ductility markedly reduced owing to the loss of coherency between the κ-carbides and austenite matrix and the generation of coarse intergranular κ-carbides forming a network along the austenite grain boundaries which promote the nucleation of cracks [12]. The formation of brittle β-Mn at grain boundaries when the Mn concentration exceeds 30 wt% during prolonged aging has also been reported to be detrimental to the ductility and toughness, contributing to the brittle fracture and loss in the impact energy [13]. Moreover, the shearable coherent κ-carbides are prone to result in strain softening effect and thus contribute to the strain localization once they are cut by the moving dislocations during plastic deformation. The coarsening of the κ-carbides during the extended aging process further aggravates the strain softening leading to a significant decrease in strain-hardening capability [14]. As a result, the age-hardening process of the Fe-Mn-Al-C low-density steels must be cautiously controlled to avoid the abovementioned side effects.

Hot rolling has been reported to be a feasible approach for strength

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enhancement by increasing the rolling reduction ratio, which is mainly thanks to the grain-refinement effect arising from the steady dynamic recrystallization (DRX) phenomenon [15,16]. Extensive research regarding the DRX behavior during high-temperature plastic deformation has been carried out. The DRX kinetic model for the hot-rolled condition proposed by Peng et al. revealed that the extent of DRX was closely related to the hot-deformation conditions, such as deformation temperature and strain rate [15]. Zhu et al. [16] investigated the DRX behavior of a medium carbon Cr-Ni-Mo-Nb steel during hot deformation. The average grain size of the samples after DRX could be gradually refined by introducing larger stored energy, e.g., increasing the strain rate from 0.1 to 5 s^{-1}, which promoted the formation of critical nuclei for recrystallization. Based on the grain refinement effect introduced by DRX during hot deformation, the austenitic low-density steels could be strengthened by modifying the hot-deformation parameters.

In this study, the hot-rolled low-density austenitic steel with a chemical composition of Fe-31Mn-9Al-1C-0.45Nb-0.45Mo-0.45 W was applied to investigate the variation of microstructures and mechanical properties at different hot-rolling reduction ratios. The contributions of the various mechanisms in terms of strengthening and strain-hardening to the resultant mechanical properties have been systematically conducted, which has not been reported previously. The results of this work may provide a feasible approach to enhance the overall mechanical properties of low-density steel with fully austenitic microstructure.

2. Applied methods

2.1. Experimental

In the present study, the investigated Fe-Mn-Al-C low-density steel possessed a nominal chemical composition of Fe-31Mn-9Al-1C-0.45Nb-0.45Mo-0.45 W (wt%). The exact content of alloying elements (wt%) was: Mn 30.9, Al 8.7, C 0.9, Nb 0.45, Mo 0.45, W 0.43, and Fe in balance. The steel was fabricated using the high-frequency vacuum induction melting furnace and then hot-forged into billets with a thickness of 110 mm at 1150 °C and then air cooling to room temperature. The hot-forged billets were initially reheated at 1150 °C and then hot-rolled into plates with thicknesses of 60 mm, 35 mm, and 20 mm (hot-rolling reduction: 45%, 68%, 82%), which were denoted as HR45, HR68, and HR82 samples, respectively, and then water quenching to room temperature. The number of passes during hot rolling were 4, 6, and 9 for the HR45, HR68, and HR82 samples. The density of the investigated steel is 6.82 g/cm^3 (measured using Archimedes' principle). In comparison to pure iron, a density reduction of ~13% was attained.

The hot-rolled microstructures of the HR45, HR68, and HR82 samples were characterized by the scanning electron microscope (SEM) and electron back-scattered diffraction (EBSD). The SEM and EBSD analyses were performed in a JSM-7200F scanning electron microscope with a backscattered electron detector at 20 kV. The original data obtained from EBSD with a step size of 1 μm were analyzed by the HKL channel 5 software. The mean grain sizes for the grains with minimum misorientation angles of 15° (excluding annealing twin boundaries) were determined using the line intercept method from the EBSD data. The boundaries with a misorientation angle of 60 ± 0.5° were identified as twin boundaries. The mean sizes of the precipitates were calculated using the equation as follows from the SEM or TEM micrographs:

$$f = Vn/2Sd \tag{1}$$

where f represents the precipitates volume fraction, V is the average volume of precipitates calculated as spheres, n is the number of precipitates in the measured area, S is the mean size of the measured areas, and d reflects the mean diameters of the precipitates.

The morphology of precipitates and the deformation substructures were investigated using the transmission electron microscopy operated at 200 kV (TEM, FEI Tecnai G2 F20), and the identification of the chemical composition of the precipitates was carried out using the energy-dispersive X-ray spectroscopy (EDS) attached to the TEM. The standard tensile samples were cut from the hot-rolled plate samples along the rolling direction (RD) with a gauge length of 25 mm and a diameter of 5 mm.

The uniaxial tensile tests were conducted on an MTS tensile machine at an initial strain rate of 1.0×10^{-3} s^{-1} to investigate the stress-strain responses and strain-hardening behavior of HR45, HR68, and HR82 samples. The extensometer was applied for obtaining more accurate results. To further examine the evolution of deformation substructures during plastic deformation, the tensile test was interrupted at the engineering strain of 10%, 20%, and 40% before fracture. The impact toughness of the samples at different hot-rolling reductions was determined by the Charpy V-notch impact test on the standard samples (55 mm in length, 10 mm in width, 10 mm in thickness, and a V-notch depth of 2 mm) at room temperature, of which the length was parallel to the rolling direction and the striking direction of the impact test was parallel to the transverse direction.

The stacking fault energy (SFE) of the present steel Γ was calculated based on the thermodynamic model proposed by Olson and Cohen [17]:

$$\Gamma = 2\frac{4}{\sqrt{3}a^2N_A}\Delta G^{\gamma\to\varepsilon} + 2\sigma \tag{2}$$

where a is the lattice constant of austenite, N_A is the Avogadro constant, $\Delta G^{\gamma\to\varepsilon}$ is the free energy difference between the γ-phase and ε-phase, σ is the γ/ε interface energy.

3. Results

3.1. Examination of microstructure at different hot-rolling reductions

The microstructures of the investigated samples at different rolling reductions are illustrated with EBSD maps, as shown in Fig. 1After hot-rolling and subsequent water quenching, the microstructures of all the samples consisted of equiaxed FCC austenite grains with a large fraction of the first-order annealing twins (⟨1 1 1⟩ 60° ∑3-grain boundaries) identified inside (denoted as red lines). The mean grain sizes of the austenite phase were measured as about 45.11, 27.02, and 18.80 μm for HR45, HR68, and HR82 samples, respectively. When the annealing twin boundaries were included, the mean effective grain sizes were notably reduced to about 21.82 μm, 11.55 μm, and 7.02 μm for HR45, HR68, and HR82 samples, respectively. The results showed that increasing hot-rolling reduction gave rise to significant grain refinement and the annealing twin boundaries could effectively reduce the mean grain size of the austenite phase. The statistics of the ∑3 twin boundaries were measured to be 67%, 60%, and 52% (determined by length relative to the total length of the grain boundaries with misorientation angle >15°) for HR45, HR68, and HR82 samples, respectively.

In order to show the dislocation distribution in the investigated samples, the Kernel average misorientation (KAM) values were calculated from the EBSD-raw data, and the KAM maps are shown in Fig. 1(g-i). It is evident that the HR82 sample exhibited more areas with high orientation gradients than the other samples, suggesting that the local strain concentration due to the accumulation of dislocations was more prevalent with increasing hot-rolling reduction. The quantitative evaluation of the geometrically necessary dislocations (GND) densities in the present study was based on the approach following Kubin and Mortensen [18], who considered a GND array as simple cylinder torsion according to the strain gradient model of Gao et al. [19]. As defined, there was a series of twist boundaries in the cylinder, and each twist boundary consisted of two perpendicular arrays of screw dislocations, following which the relationship between GND density and misorientation angle θ was developed,

$$\rho^{GND} = 2\theta/ub \tag{3}$$

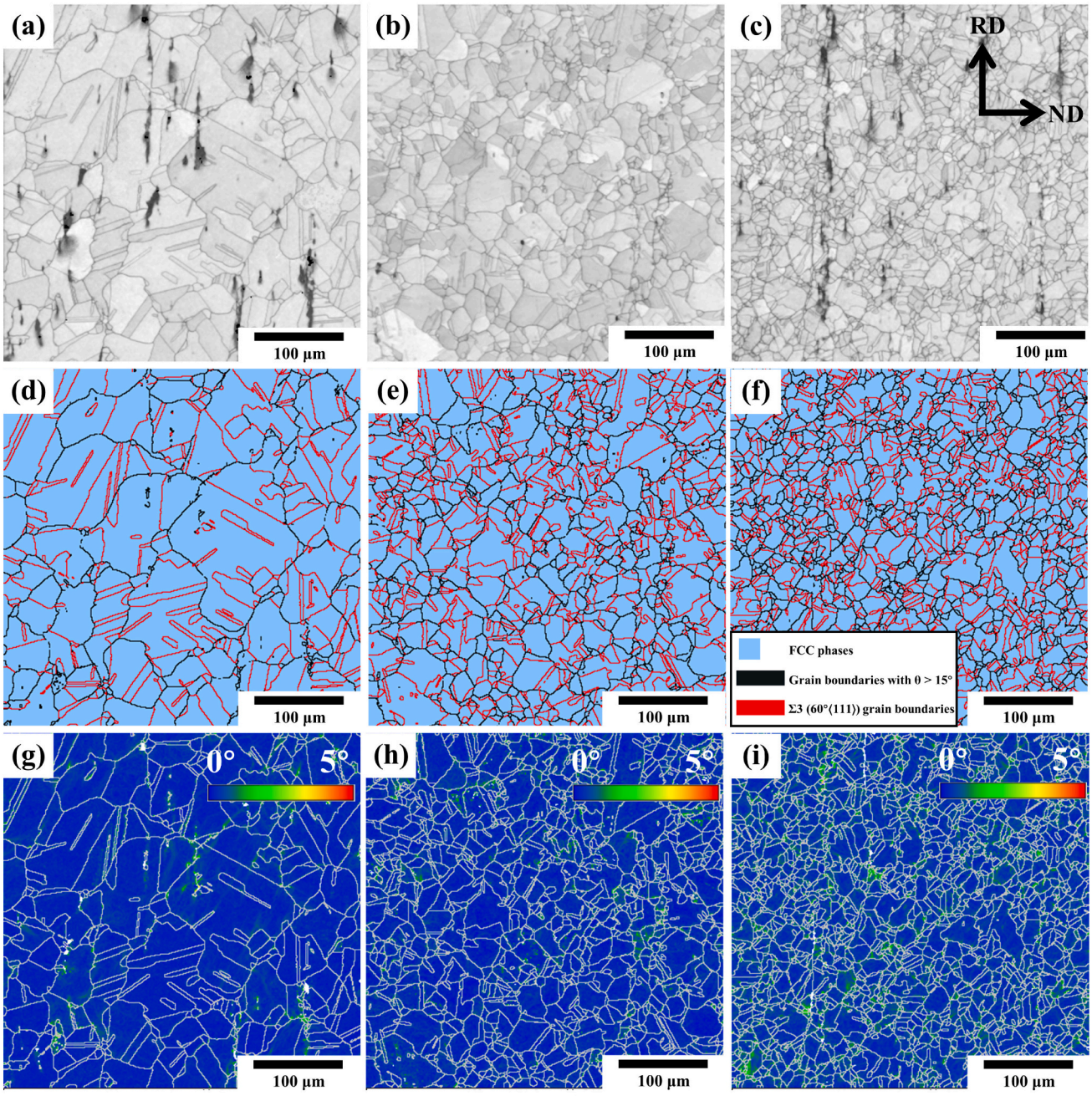


Fig. 1. EBSD (a–c) band contrast (BC) maps (d–f) phase maps overlaid by grain boundary maps and (g-i) KAM maps of the hot-rolled (a, d, g) HR45, (b, e, h) HR68, and (c, f, i) HR82 samples. The face-centered cubic (FCC) phase is shown in blue. RD and ND represent the rolling direction and normal direction, respectively. Black and red lines denote high-angle grain boundaries (HAGBs) with misorientation ≥ 15° and Σ3 (60°⟨111⟩) grain boundaries, respectively. The KAM angle ranges from 0° to 5°. The areas with high orientation gradients indicated the local strain concentration. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where u is the unit length and b is the magnitude of the Burgers vector. As a first-order approach, the KAM values obtained from the EBSD-raw data were assumed to be a representation of the local misorientations. Based on this method, the calculated mean GND densities were 2.71×10^{13} m^{-2}, 7.49×10^{13} m^{-2}, and 9.87×10^{13} m^{-2} for HR45, HR68, and HR82 samples, respectively. With increasing rolling reduction, higher mean GND densities were obtained in the measured microstructures.

As revealed by SEM in Fig. 2, some second-phase precipitates existed in the microstructures of the investigated samples after hot rolling. The micron-sized polygonal ones were distributed along the rolling direction in the grain boundary and grain interior with sizes ranging from 1 to 25 μm. The submicron-sized spherical ones were distributed homogeneously in the austenite matrix. The mean particle sizes of the submicron-sized particles were 261 ± 36 nm, 256 ± 42 nm, and 259 ± 23 nm, and the volume fractions were ~ 0.193%, ~0.201%, and ~ 0.198%, respectively for the HR45, HR68, and HR82 samples. To further evaluate the chemical compositions and the crystal structures of these precipitates, EDS analysis was applied. The crystal structure of the precipitates was identified as the face-centered cubic (FCC) structure as shown in Fig. 3, and the lattice parameter was calculated as 4.36 Å.

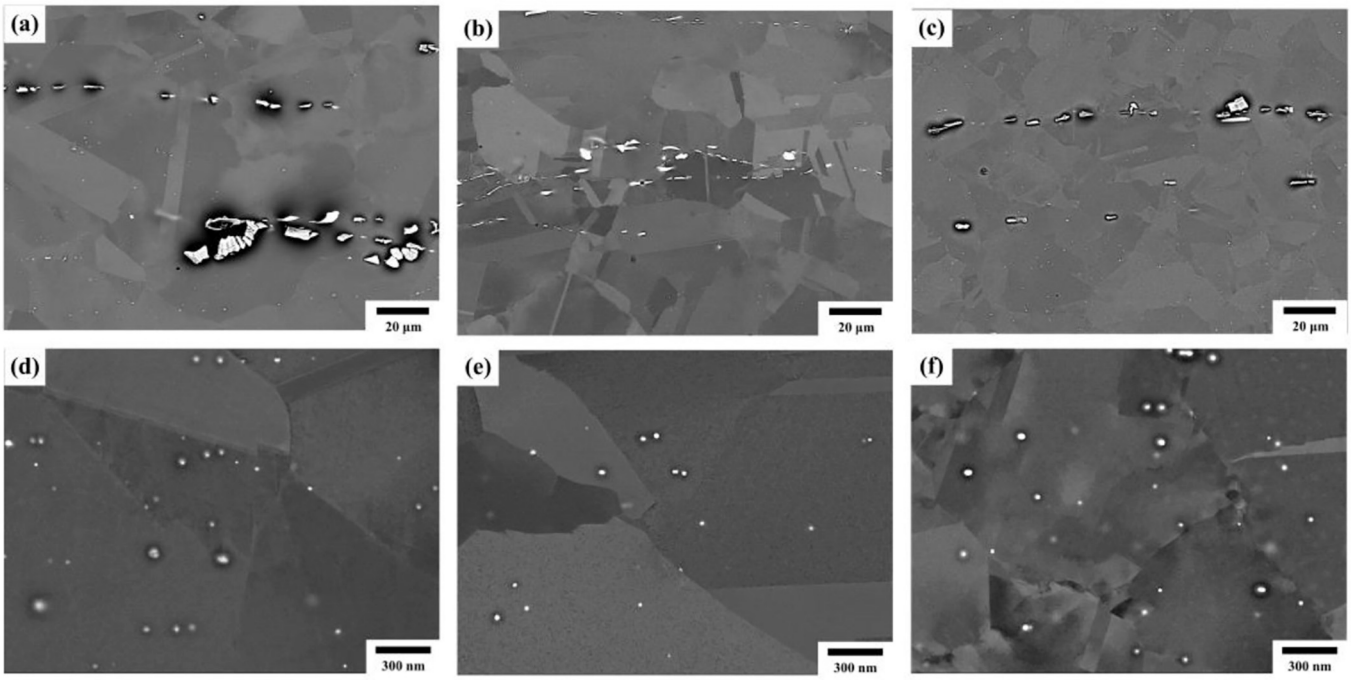


Fig. 2. SEM Backscattered-electron (BSE) image of the distribution of the (a–c) micron-sized and (d–f) submicron-sized (at higher magnification) second phase particles of the (a, d) HR45, (b, e) HR68, (c, f) HR82 samples.

Thus, they were demonstrated to be the (Nb,Mo,W)C as a class of MC-type complex carbides.

Fig. 4(a-c) shows the selected-area electron diffraction (SAED) patterns taken from the austenite matrix of the three investigated samples along the [1 0 0]-zone axis. The faint (1 0 0) and (1 0 1) superlattice reflections indicated the existence of the second phase with an ordered crystal structure. The corresponding TEM-dark field (DF) images taken from the same areas using the (1 0 0) superlattice reflection are shown in Fig. 4(d-f), demonstrating that a large number of nano-sized particles homogeneously precipitated inside the austenite grains during water quenching in all the samples. According to the similar observation reported in other Fe-Mn-Al-C alloys, these ordered finely dispersed particles were hence identified as the $(Fe,Mn)_3AlC$-type κ-carbides with an $L'1_2$-type structure, which was produced by the spinodal decomposition and subsequent ordering reaction during fast cooling without the further aging process [20–22]. The mean sizes of the κ-carbides were 0.81 ± 0.26 nm, 0.78 ± 0.17 nm, and 0.83 ± 0.23 nm for the HR45, HR68, and HR82 samples, respectively. The weak diffraction spots indicated the low volume fractions of the κ-carbides, which were 3.5 ± 0.33%, 3.3 ± 0.52%, and 3.9 ± 0.39% for the HR45, HR68, and HR82 samples, respectively.

3.2. Mechanical properties

The room-temperature engineering stress (σ_e)-strain (ε_e) curves of the investigated samples at different hot-rolling reductions are illustrated in Fig. 5a, from which the mechanical properties are summarized in Table 1. The HR45 sample with a 45% rolling reduction exhibited an ultimate tensile strength (UTS) of 861 ± 11 MPa, a yield strength (YS) of 480 ± 0 MPa, and a total elongation of 60%. The increase in rolling reduction gave rise to the enhancement in both the UTS and YS and a minor loss of uniform elongation (UE) and total elongation (TE). With the increase in rolling reduction from 45% to 82%, the improvement of UTS and YS reached ~110 MPa, and ~ 170 MPa, respectively, while the UE decreased from 52.0 ± 1.0% to 39.0 ± 0.8, and TE decreased from 60 ± 1% to 50 ± 1%. In addition, all three samples possessed a similar reduction of area of >60%, which suggested their excellent ductility. The impact toughness is the capability of energy absorption until fracture, which can be manifested by the Charpy V-notch (CVN) impact energy, as exhibited in Fig. 5b. All the samples at different hot-rolling reductions possessed outstanding Charpy impact energy above 100 J/m^2. The increase in hot-rolling reduction from 45% to 82% caused a minor loss of the Charpy impact energy from 133 ± 2.5 J/m^2 to 105 ± 1.2 J/m^2, suggesting good toughness was maintained.

The fracture surfaces of the investigated samples after the tensile test were examined using SEM, as exhibited in Fig. 6. The fractography of all the hot-rolled samples shows a typical ductile fracture with a combination of ductile dimples and various microvoids. In general, the (Nb,Mo,W)C particles located at the austenite grain boundaries (see Fig. 2) as a hard second phase particle will lead to the strain/stress concentration at their interfaces due to their deformation discrepancy with the austenite matrix. Thus, they tend to severed as the preferential nucleation sites for the microcracks. Nevertheless, the strain/stress concentration at the interfaces of these intergranular (Nb,Mo,W)C particles could be partially alleviated through the debonding of the intragranular (Nb,Mo,W)C particles and the subsequent formation of microvoids (see Fig. 6) [23,24]. Considering the abundantly observed ductile dimples on the fracture surface of the three investigated samples, the intergranular (Nb,Mo,W)C particles virtually played no detrimental effect on the toughness. With the increase in the rolling reduction, the ductile fracture morphology was retained, suggesting the toughness of the investigated samples was barely affected, which could also be verified by the Charpy impact test results (Fig. 5b). In addition, the size of the dimples in the HR45 sample was larger than the HR68 and HR82 samples, which was due to its larger mean grain size [25].

Fig. 7 shows the yield strength, total elongation, and Charpy impact energy of the HR82 sample and those of other austenitic steel (tested on the standard impact specimen at room temperature). In comparison to the austenitic stainless steels and the Fe-Mn-Al-C alloys at the solution-treated state [26], the HR82 sample exhibited a similar level of ductility (~50%) and toughness (~100 J/m^2) while the yield strength was markedly higher. The yield strength of the HR82 sample is comparable to the aging-hardened austenitic low-density steel (~650 MPa) [27,28] but possessed considerably better ductility (50% vs. 21%) [27] and

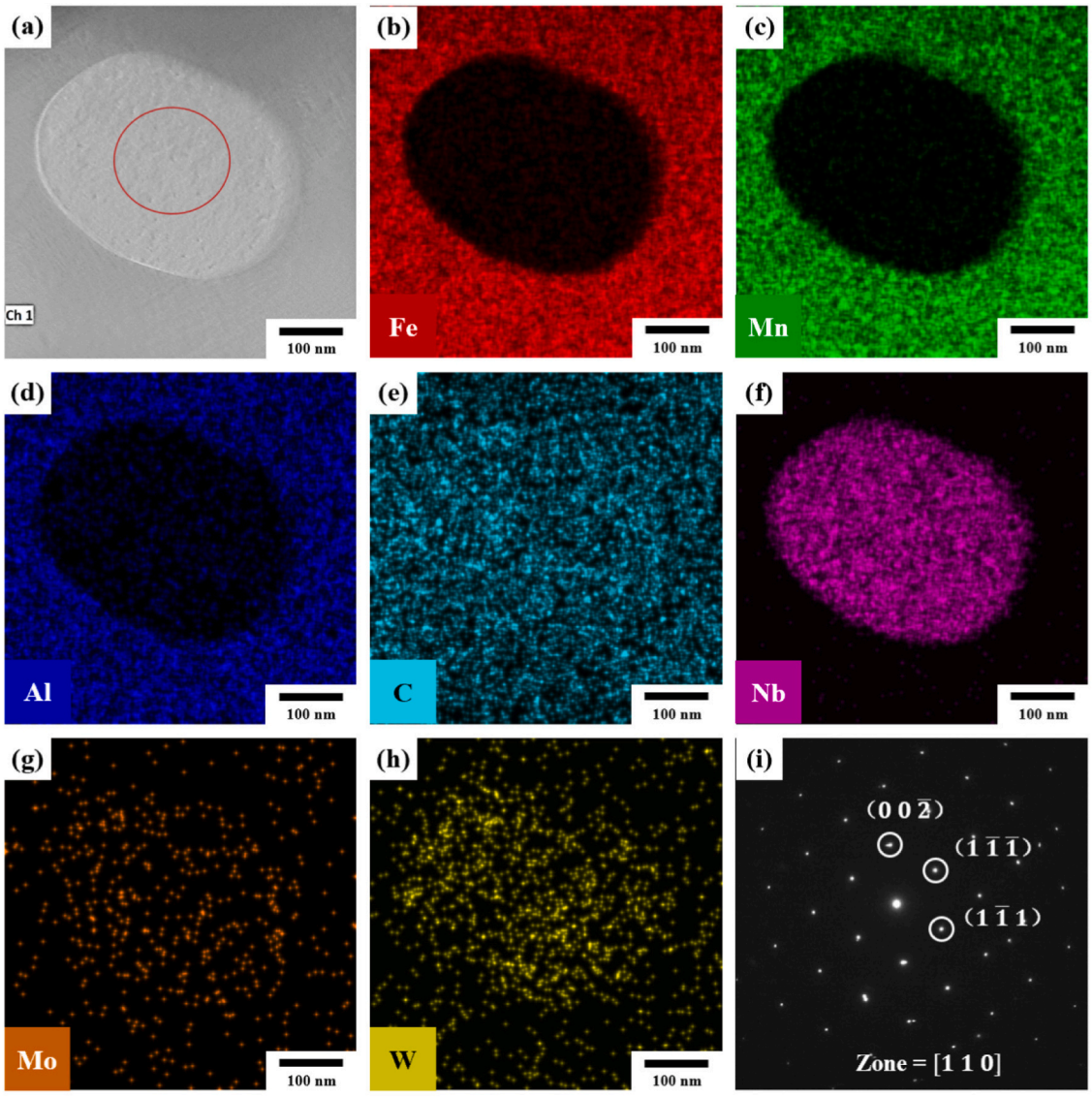


Fig. 3. (a) TEM-bright field (BF) images, and corresponding EDS maps of (b) Fe, (c) Mn, (d) Al, (e) C, (f) Nb, (g) Mo, (h) W of the submicron-sized particle in the HR68 sample (i) corresponding selected-area electron diffraction (SAED) patterns of the [110]-zone axis taken from the submicron-sized particle.

toughness (105 J vs. 35 J) [28]. As a result, the distinct feature of the present steel is an excellent combination of strength, ductility, and toughness obtained by increasing the hot-rolling reduction from 45% to 82%.

4. Discussion

4.1. Effect of hot-rolling reduction on the microstructure and mechanical properties

As shown in Fig. 1(a-c), the increase in hot-rolling reduction resulted in the refinement of the grain size. This was mainly attributed to the enhanced dynamic recrystallization (DRX) for the sample subjected to larger rolling reduction. Additionally, the intenser particle-stimulated nucleation (PSN) might be another influencing factor considering the existence of a large number of (Nb,Mo,W)C particles (Fig. 2). Besides, the deformation-induced dislocation accumulation during hot-rolling led to the continuous increase in geometrically necessary dislocations (GND) density, as shown in Fig. 1(g-i). The parameters of the grain size and dislocation density are revealed in Fig. 8(a) for the understanding of the strengthening mechanism, i.e., grain boundary strengthening and dislocation strengthening. Furthermore, as shown in Fig. 2 and Fig. 4 the presence of the (Nb,Mo,W)C particles and κ-carbides would give rise to the precipitate strengthening as well. Fig. 8(b) shows the volume fractions and mean particle sizes of the (Nb,Mo,W)C particles and κ-carbides of the three investigated samples. As listed in Table 1, with the increasing hot-rolling reduction from 45% to 82%, the ultimate tensile strength (UTS) and yield strength (YS) gradually increased from 861 ± 11 MPa and 480 ± 0 MPa to 976 ± 11 MPa and 648 ± 13 MPa, respectively. To reveal the strengthening factors that resulted in the increment of yield strength with increasing hot-rolling reduction, the contribution of the different strengthening mechanisms will be discussed and calculated.

According to the strengthening mechanisms mentioned above, the YS could be determined as follows:

$$\sigma_y = \sigma_0 + \sigma_p + \sigma_{ds} + \sigma_{gb} \quad (5)$$

where σ_y is the yield strength, σ_0 is the friction stress of the austenite matrix (including the solid solution strengthening), σ_p is the precipitate strengthening contribution, σ_{ds} is the stress arising from the dislocation strengthening, and σ_{gb} is the stress due to the grain boundary strength-

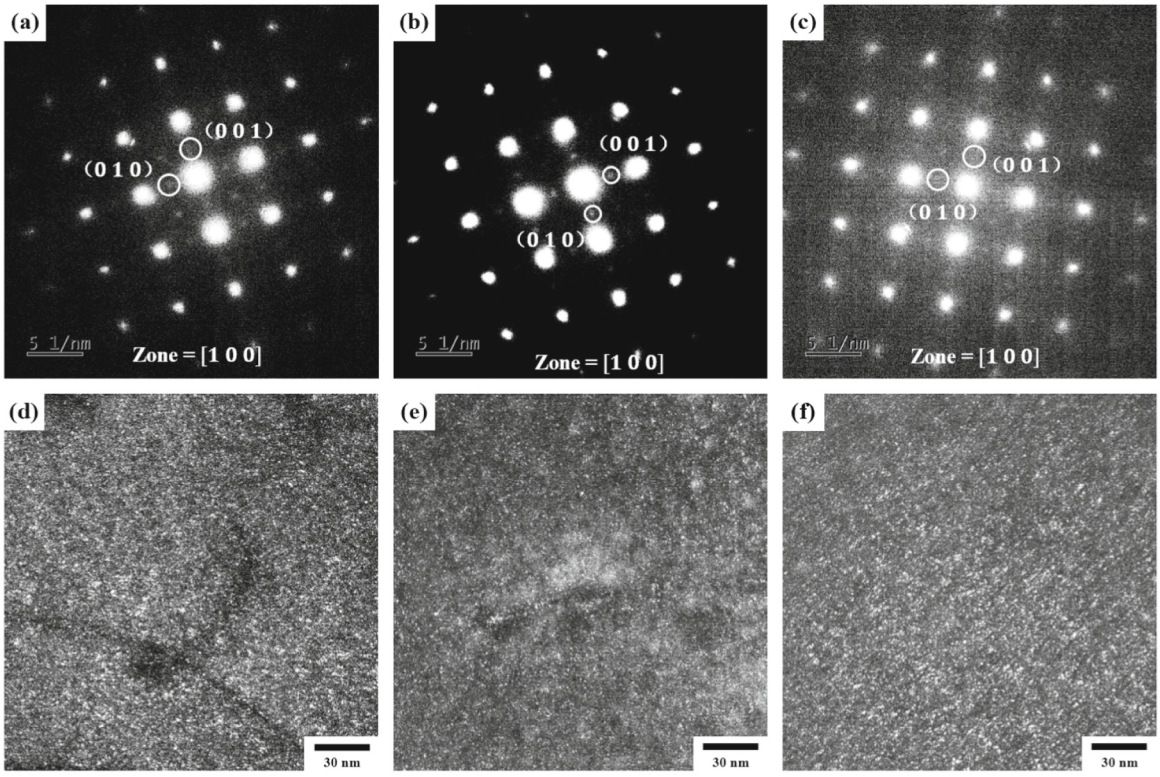


Fig. 4. κ-carbide TEM images: (a–c) the SAED patterns taken from the austenite matrix of the (a, d) HR45, (b, e) HR68, (c, f) HR82 samples along the [1 0 0]-zone axis, and (d–f) the corresponding dark field (DF) images using the (1 0 0) superlattice diffraction taken from the same areas. The faint superlattice diffraction spot marked with circles indicated the presence of κ-carbides.

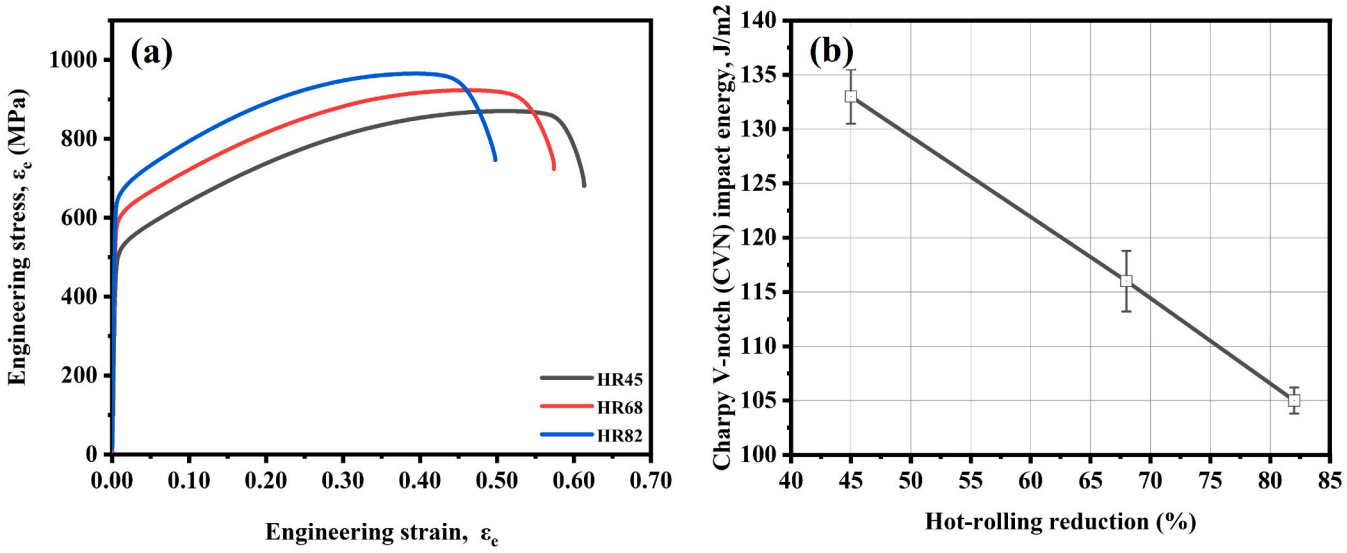


Fig. 5. Mechanical properties of the investigated samples at different hot-rolling reductions (a) engineering stress (σ_e)-strain (ε_e) curves, (b) Chary V-notch (CVN) impact energy as a function of hot-rolling reduction.

Table 1
The mechanical properties of the investigated samples at different hot-rolling reductions.

Rolling states	Ultimate tensile strength, MPa	Yield strength, MPa	Uniform elongation, %	Total elongation, %	Reduction of area, %	CVN impact energy, J/m^2
HR45	861.0 ± 11.0	480.0 ± 0.0	52.0 ± 1.0	60.0 ± 1.0	63.0 ± 1.0	133 ± 2.5
HR68	922.0 ± 2.0	577.0 ± 0.0	47.0 ± 0.5	57.0 ± 0.0	63.0 ± 0.0	116 ± 2.8
HR82	976.0 ± 11.0	648.0 ± 13.0	39.0 ± 0.8	50.0 ± 1.0	61.5 ± 0.3	105 ± 1.2

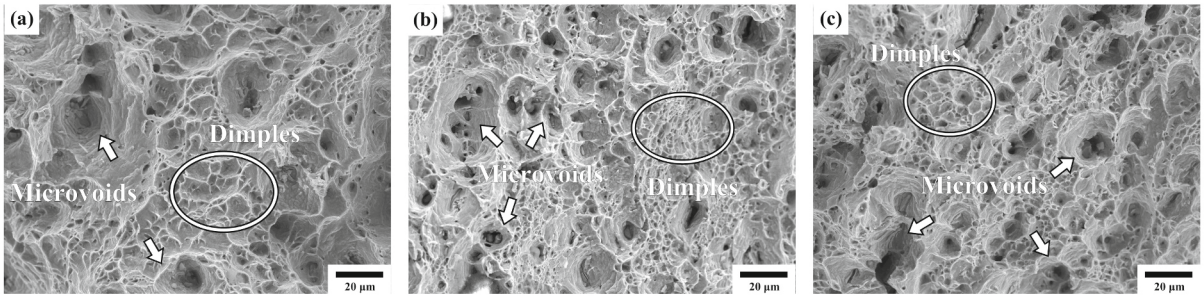


Fig. 6. SEM fractography of the (a) HR45, (b) HR68, and (c) HR82 samples, dimples and microvoids were denoted by the circles and arrows, respectively.

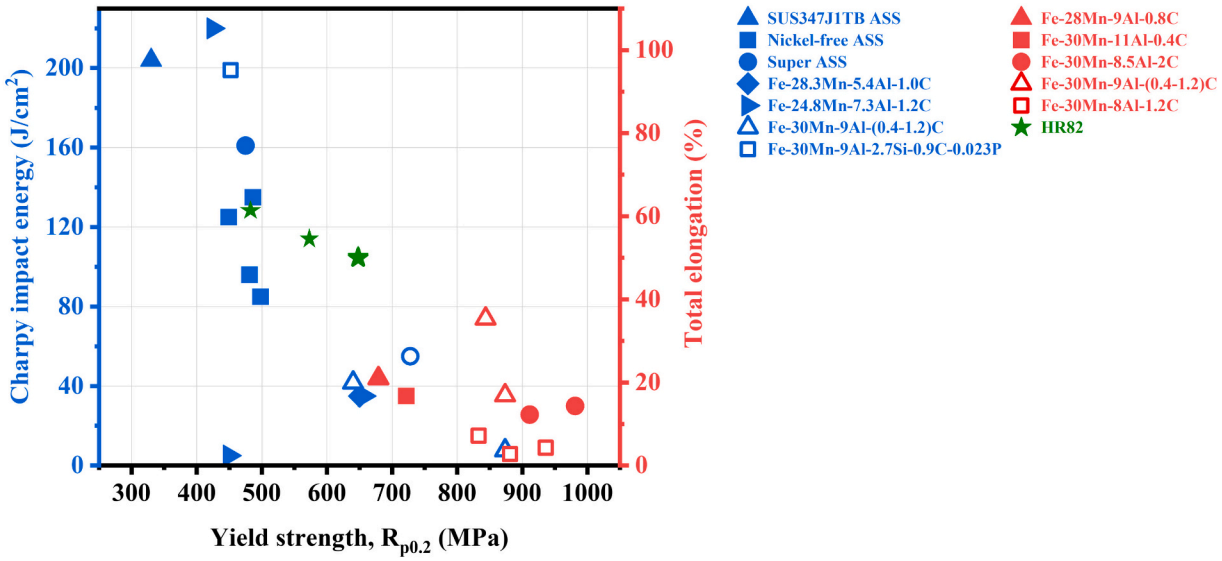


Fig. 7. Mechanical properties of the HR82 sample compared to other austenitic steels: SUS347J1TB austenitic stainless steels (ASS) [29], Nickel-free austenitic stainless steels (ASS) [26], Super austenitic stainless steels (ASS) [30], and other Fe-Mn-Al-C low-density steels [26–28,31–36].

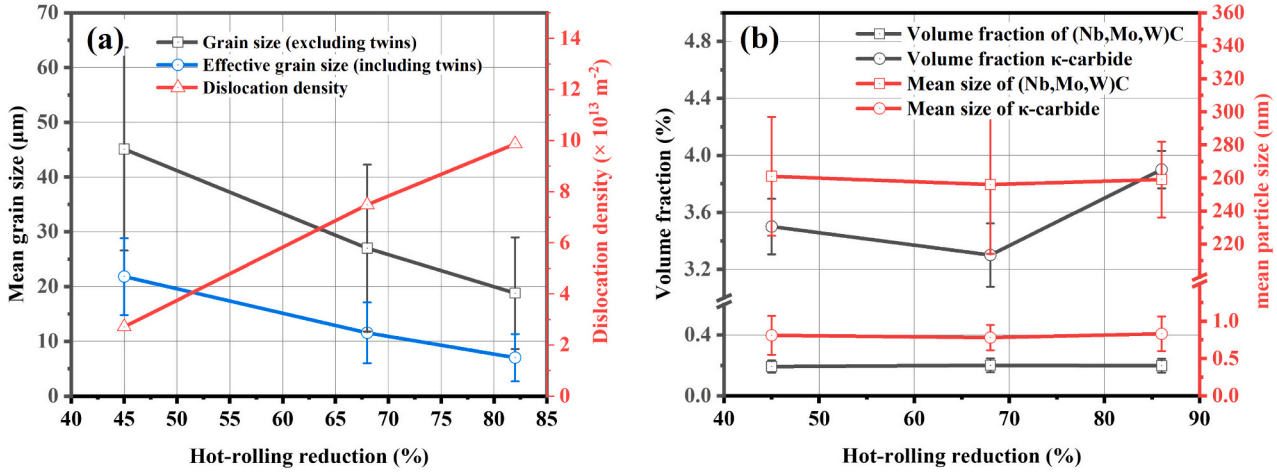


Fig. 8. (a) The variation of the mean grain size excluding and including twin boundaries and the dislocation density of the samples at different hot-rolling reductions, and (b) the variation of volume fractions and mean particle sizes of the κ-carbides and (Nb,Mo,W)C particles of the three investigated samples.

ening.

Since the chemical composition of the three investigated samples was the same, lattice friction stresses were regarded to be equivalent. The precipitate strengthening contribution was mainly attributed to the strengthening from the interaction of dislocations with the κ-carbides and the (Nb,Mo,W)C particles. Considering the particle size and co-

herency with the austenite matrix, the dislocations could shear the κ-carbides [14,37] but were only able to bypass the (Nb,Mo,W)C particles through the Orowan mechanism during plastic deformation [38,39]. The shearing mechanism between gliding dislocations and the κ-carbides generates the anti-phase boundaries (APBs), which require the following dislocations to remove the APBs, contributing to the coupling of dislocation pairs. Considering the extremely small size (< 1 nm) of the κ-carbides in the investigated samples, the coupling between dislocation pairs was weak [14]. Therefore, the strengthening contribution of the κ-carbides was determined by:

$$\sigma_{p-\kappa} = M\frac{\gamma_{APB}}{2b}\sqrt{f}\cdot\left(\sqrt{\frac{12\gamma_{APB}r}{\pi Gb^2}} - \sqrt{f}\right) \tag{6}$$

where M represents the Taylor factor of FCC alloys (3.06), G is the shear modulus, was taken as 68 GPa, b is the magnitude of the Burgers vector (0.26 nm), f reflects the volume fraction of the κ-carbides, r is the mean radius of the κ-carbides, γ_{APB} is the APB energy, estimated as ~350 mJ/m^2.

The strengthening contribution of the (Nb,Mo,W)C particles through the Orowan by-passing mechanism follows the Eq. [40]:

$$\sigma_{p-(\mathrm{Nb,Mo,W})\mathrm{C}} = M\frac{0.4Gb}{\pi\sqrt{1-\nu}}\frac{ln(2r/b)}{2r\left(\sqrt{\frac{\pi}{4f}}-1\right)} \tag{7}$$

where M represents the Taylor factor of FCC alloys (3.06), G is the shear modulus (170 GPa), b is the magnitude of the Burgers vector (0.26 nm), ν is Poisson's ratio, r is the mean radius of the (Nb,Mo,W)C particles, f is the volume fraction of the (Nb,Mo,W)C particles.

The volume fractions and mean particle sizes of the κ-carbides and (Nb,Mo,W)C particles of the three investigated samples are shown in Fig. 8(b). According to Eq.6, the strengthening contributions of the κ-carbides by the shearing mechanism were 60.16 MPa, 58.04 MPa, and 60.98 MPa for the HR45, HR68, and HR82 samples, respectively. Owing to their low volume fraction and the large size of the micron-sized (Nb,Mo,W)C particles, their strengthening contribution via Orowan by-passing mechanism was negligible. Based on Eq.7, the strengthening from the submicron-sized (Nb,Mo,W)C particles was calculated as 6.39 MPa, 6.65 MPa, and 6.52 MPa for the HR45, HR68, and HR82 samples, respectively. According to the calculated results above, it is reasonable to conclude that the precipitate strengthening did not contribute to the strength increment of the investigated samples with the larger hot-rolling reduction.

The contribution of dislocation strengthening was attributed to the geometrically necessary dislocations (GNDs). The calculation results from the KAM maps (Fig. 8(a)) suggested that the mean GND density rose with the larger rolling reduction, since the unrelieved deformation energy after hot rolling was higher with the larger rolling reduction. The GND densities obtained from the KAM maps as already shown in Section 3.1 were 2.71×10^{13} m^{-2}, 7.49×10^{13} m^{-2}, and 9.87×10^{13} m^{-2} for the HR45, HR68, and HR82 samples, respectively. The strengthening contribution of the dislocations was calculated by the Taylor hardening law [41]. Correspondingly, the strengthening arising from the dislocations was 56.29 MPa, 93.67 MPa, and 107.50 MPa for HR45, HR68, and HR82 samples, respectively.

$$\sigma_{ds} = M\alpha Gb\rho_{ds}^{1/2} \tag{8}$$

where M reflects the Taylor factor of FCC alloys (3.06), α is a numerical constant (0.2), G is the shear modulus (68 GPa), b is the magnitude of the Burgers vector (0.26 nm), and ρ_{ds} is the residual dislocation density.

The grain boundary strengthening was ascribed to the barrier effect of grain boundaries to the dislocation movement during plastic deformation. In addition, the twin boundaries also could act as effective impedance sites to the motion of dislocations and reduce their mean free path, which plays a similar role as the HAGBs [42]. The effective grain size including annealing twin boundaries was applied in the present study when considering the grain boundary strengthening, which generally followed a Hall-Petch relationship. The dependence of the true flow stress at yielding σ_{F0} (obtained by excluding the contributions of precipitate strengthening σ_p and dislocation strengthening σ_{ds} from the yield strength σ_y) as a function of the inverse square root of the mean effective grain size ($d_e^{-1/2}$) was plotted in Fig. 9(a). According to the fitting result, the friction stress (σ_0) of 203.10 MPa and the Hall-Petch coefficient (k_{HP}) of 733.43 MPa μm$^{1/2}$ were obtained from the y-axis intercept and slope, respectively. Hence, the contribution of grain boundary strengthening for HR45, HR68, and HR82 samples was calculated as 157.02 MPa, 215.84 MPa, and 276.78 MPa, respectively.

$$\sigma_{gb} = k_{HP}d_e^{-\frac{1}{2}} \tag{9}$$

where k_{HP} represents the Hall-Petch coefficient, and d_e is the mean effective grain size (μm).

The contributions of the various strengthening mechanisms to the yield strength of the three investigated samples were shown in Fig. 9(b) and the calculated results were summarized in Table 2, showing good consistency with the experimental results and thus demonstrating the validity of the analysis of the strengthening mechanisms. As shown in Fig. 9(b), the grain boundary strengthening arising from both HAGBs and annealing twin boundaries was the predominant strengthening mechanism and the strengthening effect from the dislocations was also prominent. They were responsible for the increment in yield strength with increasing hot-rolling reduction.

4.2. Correlation between the three-staged strain-hardening behavior and evolution of the deformation substructures

To reveal the mechanism contributing to the improved strength-ductility combination of the present steel after hot rolling, the strain-hardening rate ($d\sigma/d\varepsilon$) vs. true strain (ε_t) curves of the investigated samples were exhibited in Fig. 10. The evolution trend of the strain-hardening rate of the three investigated samples was similar, which exhibited an initial drop after yielding (stage I) followed by a gradual increase to their peak values (stage II) and then a continuous decrease until fracture (stage III), indicating a three-staged strain-hardening behavior. Despite the high SFE of ~73 mJ/m^2, the evolution of the strain-hardening rate was quite different with other conventional high-SFE fcc metals [43], which show a continuously decreasing trend with increasing strain. The three-staged strain-hardening behavior of the present steel led to the strain-hardening rate being maintained at a high level of ~2000 MPa over a wide range of plastic strain, contributing to the excellent ductility. In addition, the strain-hardening rates of the present steel increased continuously with increasing hot-rolling reduction.

As widely accepted, the evolution of strain-hardening behavior is essentially governed by the development of deformation substructures. For the TRIP or TWIP steels, the similar "hump" of the stain-hardening curve, i.e., the increment stage of strain-hardening rate, was attributed to the activation of the secondary deformation mechanism [44,45]. In order to clarify the underlying mechanism resulting in the characteristic strain hardening behavior of the present steel, interrupted tensile tests at the engineering strain (ε_e) of 0.1, 0.2, and 0.4 were conducted and the corresponding deformation microstructures were characterized by TEM. The representative TEM-bright field (BF) images (taken along the ⟨1 1 0⟩ zone axis of the austenite matrix grains) are displayed in Fig. 11. The evolution pattern of the dislocation substructures of the samples at different rolling reductions was basically the same. At low strain, the arrays of planar slip bands were distributed homogeneously within the austenite matrix (Fig. 11(a, d, and g)). Moreover, the Taylor lattice formed by the intersections of two non-coplanar {1 1 1} slip traces was also observed, as indicated by the dashed lines in Fig. 11(a, d). With increasing plastic strain, the slip bands became more closely spaced

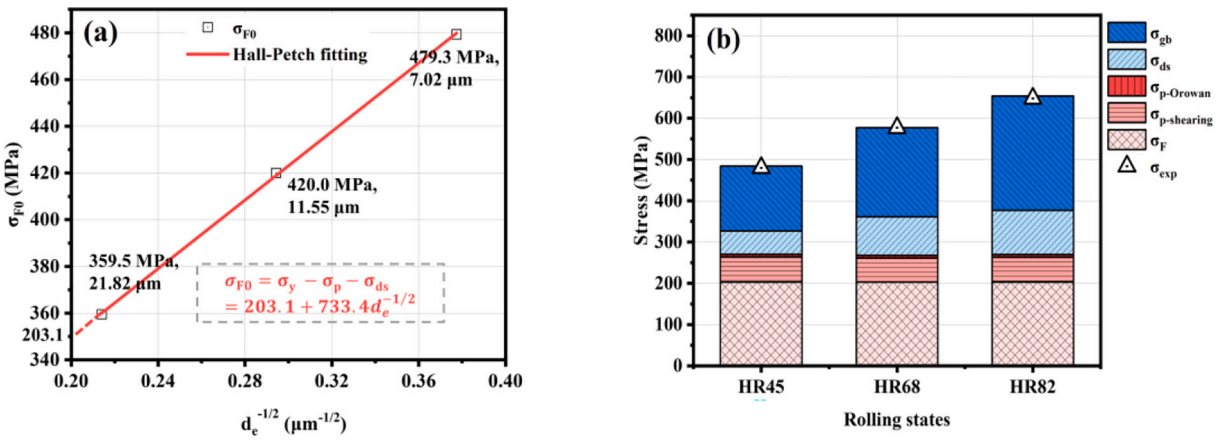


Fig. 9. (a) The dependence of the true flow stress at yielding σ_{F0} (obtained by subtracting the precipitate strengthening σ_p and dislocation strengthening σ_{ds} from the yield strength σ_y) as a function of the inverse square root of the mean effective grain size ($d_e^{-1/2}$). The Hall-Petch fitting result is denoted in the red line. (b) Theoretical calculation results of the contribution of individual strengthening mechanism to the yield strength of the investigated samples. The triangles indicate the experimentally tested results. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
The summarization of the strengthening contribution of different strengthening mechanisms to the yield strength and the experimentally tested results of the investigated samples.

Rolling states	σ_0, MPa	$\sigma_{p-\kappa}$, MPa	$\sigma_{p-(Nb,Mo,W)C}$, MPa	σ_{ds}, MPa	σ_{gb}, MPa	σ_{calc}, MPa	σ_{exp}, MPa
HR45	203.1	61.0	6.5	56.3	157.0	483.9	480.0
HR68	203.1	58.0	6.7	93.7	215.8	577.3	577.0
HR82	203.1	60.2	6.4	107.5	276.8	653.9	648.0

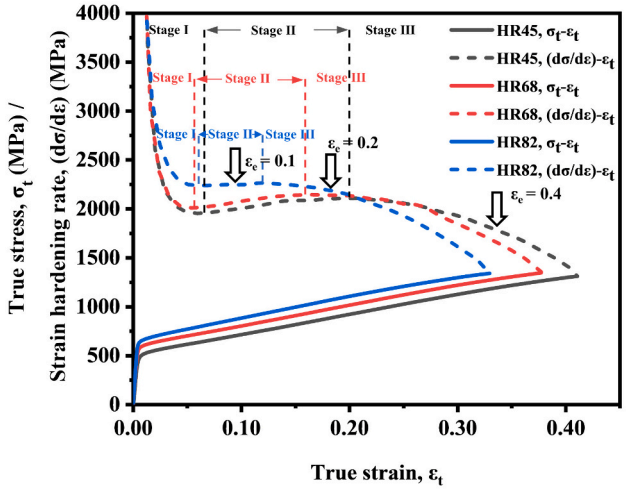


Fig. 10. True stress (σ_t)-strain (ε_t) curves (solid lines) superimposed over the strain-hardening rate ($d\sigma/d\varepsilon$)-true strain curves (dotted lines) of the investigated samples at different hot-rolling reductions. The arrows indicate at which strains the tensile tests were interrupted.

(Fig. 11(b, e, and h)), and microbands were found with ill-defined boundaries and distinct boundaries marked with arrows and solid lines (Fig. 11(c, f, h, and i)), respectively. These dislocation substructures were the typical deformation pattern formed by predominant dislocation planar glide, which is generally observed in low-SFE materials, demonstrating the low three-dimensional mobility of the dislocations. However, the SFE of the present steel was calculated to be ~73 mJ/m^2. The high SFE of the investigated steel should facilitate the cross-slip of dislocations and hence form the dislocation cell structure [46], which was against the planar glide dislocation substructures in the present steel. Gerold and Karnthaler [47] explained the dislocation planar glide behavior in the high-SFE material using the term "glide plane softening", which is arising from the interaction between dislocations and κ-carbides [14,33,42,43]. Once the κ-carbide was sheared by the leading dislocation, a preferential glide path for the subsequent dislocations would be generated due to the reduced κ-carbide effective size [48], contributing to the formation of the planar slip bands and Taylor lattice. The further increase in strain led to the rotation of the Taylor lattice. As a result, microbands delineated by dislocation walls consisting of GNDs were generated to accommodate the orientation misfit between the Taylor lattice domains [49]. Therefore, the three-staged strain-hardening behavior of the present steel should be closely related to the transition of the planar deformation substructures, which will be discussed next.

At stage I, the strain-hardening rate sharply dropped, which could be interpreted by the incompletely developed slip-band substructure. In this stage, the slip band spacing was small due to the low dislocation density. Hence the hardening effect arising from the long-range stress among the slip bands can not compensate for the initial glide plane softening effect [8,20]. The accumulation of dislocations was negligible and the dynamic recovery of dislocation was dominant [48]. The above mechanisms led to a decrease in the strain-hardening rate during stage I.

At stage II, the strain-hardening rate increased continuously. The dislocation substructures at this stage of the three investigated samples were the fully-developed slip bands across the entire grains, as shown in Fig. 11(a), (b), (d), and (g). With increasing plastic strain, the spacing among the slip bands gradually reduced. The variation of the slip band spacing with increasing plastic strain of the three investigated samples was exhibited in Fig. 12. The continuous refinement of the slip bands was attributed to the activation of dislocation sources and the formation of new slip bands in the free spaces between the fully developed ones to accommodate the increasing plastic strain. As a result, the stored dislocation density increased continuously and homogeneously. Besides, the progressively reduced slip band spacings led to increasing long-range stress among slip bands [20]. Therefore, the strengthening effect

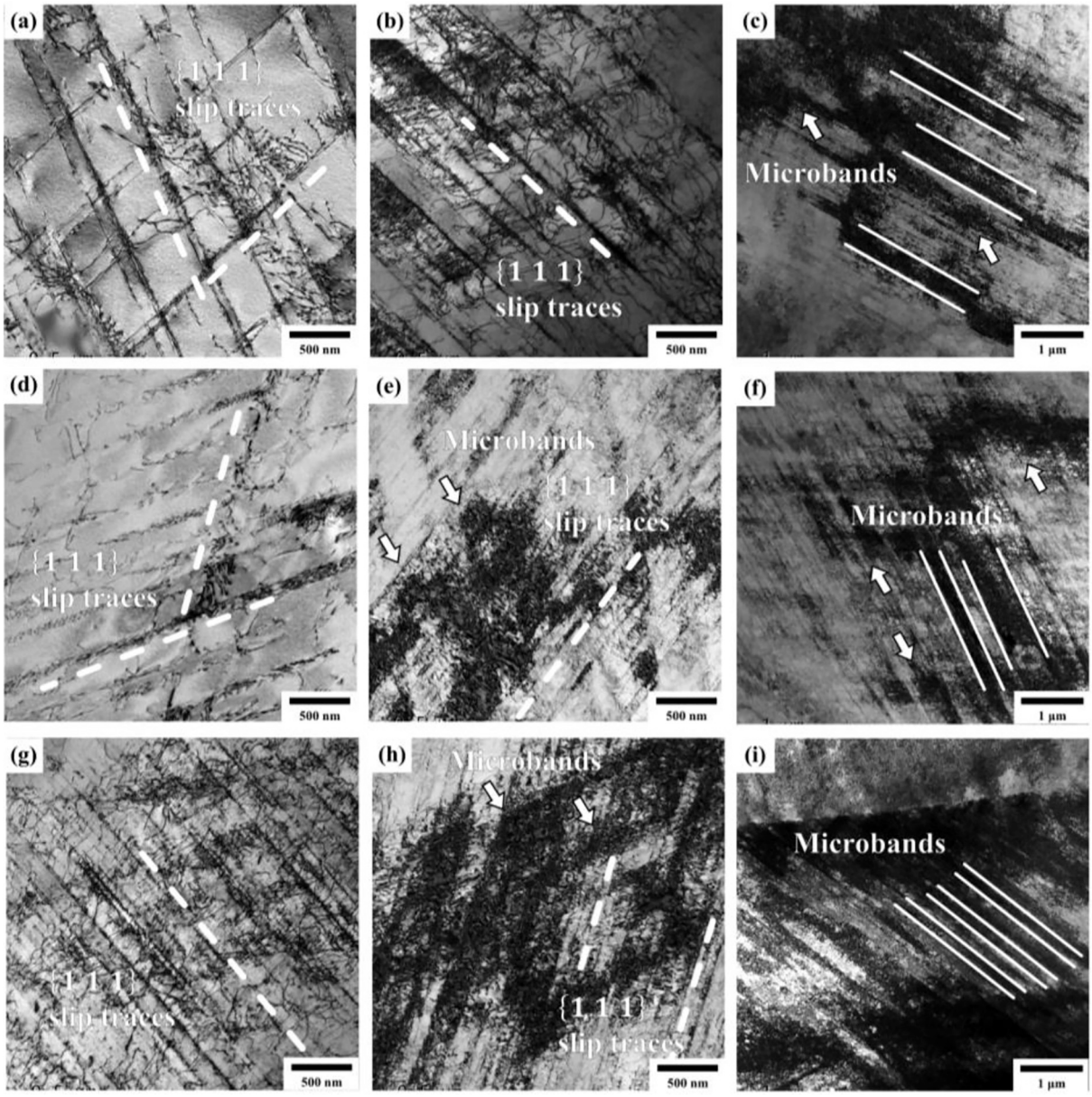


Fig. 11. TEM-bright field (BF) images of the deformation substructures taken from ⟨1 1 0⟩ zone axis of the austenite matrix of the (a–c) HR45, (d–f) HR68, (g–i) HR82 samples after (a, d, g) 0.1, (b, e, h) 0.2, (c, f, i) 0.4 engineering strain (ε_e). The coplanar and non-coplanar slip traces of the {1 1 1} slip plane were denoted by the dashed lines. Microbands with ill-defined boundaries and distinct boundaries were denoted by the arrows and solid lines, respectively.

from the above mechanisms led to the monotonously increased strain-hardening rate at stage II. It is worth mentioning that the slip band spacing barely changed after reducing to ~50 nm, which indicated that the slip band refinement process during stage II was saturated and could not provide additional strain-hardening effect by reducing the slip band spacing. The accommodation of further plastic strain should be sustained by the formation of other dislocation substructures.

At stage III, the strain-hardening rate decreased slowly and remained at a high level (~ 2000 MPa) until necking occurred. As observed in Fig. 11(c, e, f, h, and i), developing microbands with blurred boundaries and well-developed microbands with distinct boundaries were formed. This suggested that the further increase in plastic strain after the saturation of slip band refinement was accommodated by the formation and development of the microbands. The continuous development of the microbands also gradually increased the total dislocation density and subdivided the grains by their mutual intersections, i.e., the dynamic Hall-Petch effect, providing further strain-hardening effect [49]. Nevertheless, compared to the slip band refinement process which homogeneously increased the statistically stored (SSDs) dislocations and produced the long-range stress among the slip bands, the strain-hardening effect from the formation of microbands was relatively weak since they consisted of GNDs [50]. Therefore, the strain-hardening rate gradually dropped during stage III.

In addition, it is interesting to note that the evolution of deformation substructures was faster in the samples at larger hot-rolling reductions, as displayed in Fig. 12. For instance, the slip band spacing at the engineering strain of 0.1 reduced with increasing hot-rolling reduction, which was measured to be 296.40 ± 50.06 nm, 155.02 ± 38.42 nm, and 68.05 ± 8.91 nm for the HR45, HR68, and HR82 samples, respectively (Fig. 11(a), (d), and (g)). In addition, microbands were observed at the engineering strain of 0.2 in the HR68 and HR82 samples (Fig. 11(b), (e), and (h)), which indicated a faster deformation substructure development compared to the HR45 sample. Essentially, this phenomenon was attributed to the different initial grain sizes of the samples at different

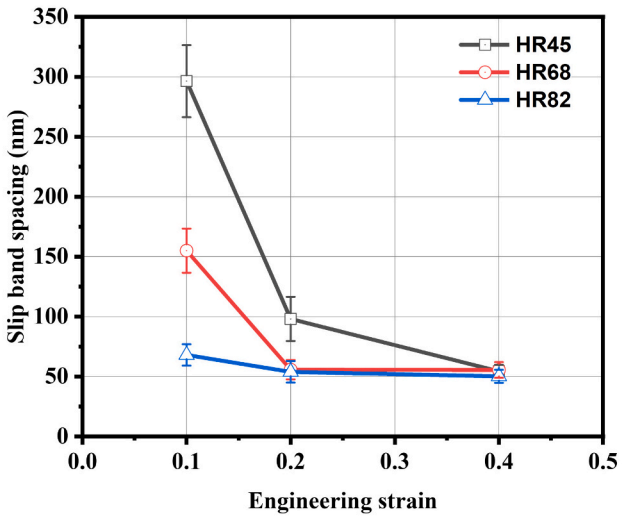


Fig. 12. The evolution of slip band spacing as a function of engineering strain of the investigated samples at different hot-rolling reductions.

hot-rolling reductions, which can be rationalized based on the dynamic slip band refinement model [20]. The glide plane softening effect leads to the dislocation multiplication on the same slip plane. The pile-up dislocations exert back stress on the dislocation sources and finally terminate the further emit of dislocations once the local stress acting on the dislocation sources falls below the critical activation stress. As shown in Fig. 1, the increase in hot-rolling reduction resulted in the finer grain size due to the enhanced DRX effect. Accordingly, the length of the slip band that generally passed through the whole grains decreased with decreasing grain size, indicating the distance between the pile-up dislocations in the slip band and the dislocation source was smaller. As a consequence, for the sample with finer grains, higher accumulated back stress acting on the dislocation source was obtained with the same amount of pile-up dislocations in the slip band. Thus, the dislocation sources will be terminated at a lower strain, suggesting the plastic strain accommodated by an individual slip band decreased with the finer grain size. The accommodation of the further plastic strain will hence require the activation of new dislocation sources, which leads to faster slip band refinement with decreasing grain size, and thus the long-range stress among slip bands was increased. This was the reason for the increased strain-hardening rates at stage II of the samples at a larger hot-rolling reduction. Nevertheless, the slip band spacing could not be further reduced once reached the limit of ~50 nm, as exhibited in Fig. 12, which suggested the saturation of the slip band refinement process. As a consequence, the faster slip band refinement contributed to the earlier saturation of this process, leading to the earlier generation of microbands and the loss of strain-hardening capability. Correspondingly, the duration of stage II decreased in the samples with finer grain sizes, which was the reason for the slight reduction of ductility.

5. Conclusions

In the present study, an excellent combination of strength, ductility, and toughness of a Fe-31Mn-9Al-0.9C-0.45Nb-0.45Mo-0.45 W (wt%) austenitic low-density steel was obtained by increasing the hot-rolling reduction from 45% to 82%. The contributions of different strengthening mechanisms and the correlation between the evolution of the deformation substructures and the strain-hardening behavior have been systematically investigated for the samples at different hot-rolling reductions. The major conclusion can be drawn as follows:

1. With the increasing hot-rolling reduction from 45% to 82%, the mean effective grain size including annealing twin boundaries was refined from 21.82 μm to 7.02 μm thanks to the enhanced dynamic recrystallization (DRX), which provided more nucleation sites during hot deformation, and the residual dislocation density after hot rolling increased from 2.71×10^{13} m^{-2} to 9.87×10^{13} m^{-2} due to the higher unrelieved deformation energy, while the volume fraction of the precipitates barely changed.
2. With the increase in the hot-rolling reduction, the mechanical properties were enhanced from the hot-rolling reduction of 45% to 82% with yield strength and ultimate tensile strength increased from 480.0 ± 0.0 MPa to 648.0 ± 13.0 MPa and 861.0 ± 11 MPa to 976.0 ± 11.0 MPa, respectively, while the ductility and the Charpy impact energy slightly decreased from 60.0 ± 1.0% to 50.0 ± 1.0% and 133 ± 2.5 J/m^2 to 105 ± 1.2 J/m^2, respectively. As a result, an excellent combination of strength comparable to that of other aging-hardened low-density austenitic steels and ductility and toughness with a similar level of ductility (~50%) and toughness (~100 J/m^2) to those of other solution-treated austenitic steels.
3. The improvement of yield strength was mainly ascribed to the stronger strengthening effect from grain boundary strengthening and dislocation strengthening. The remaining good ductility and toughness of the present steel after increasing hot-rolling reduction from 45% to 82% were mainly attributed to the maintained continuous slip band refinement process and increased strain-hardening capability at stage II during strain hardening due to the faster refinement of slip bands. In addition, the slight decrease in ductility with increasing hot-rolling reduction was due to the faster development of the deformation substructures revealed by the faster saturation of slip band refinement (slip band spacing reached ~50 nm) and earlier formation of microbands in the samples with finer grain sizes.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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